

Exercise - Solved Problems

Data Distribution, Outlier Detection & Probability — Worked Solutions

Dated: 11th August 2026

This document works through all ten end-of-chapter exercises on data distribution, central tendency, outlier detection, skewness, and probability distributions. Each problem follows the same step-by-step format used in the worked examples, with the final result highlighted in a boxed answer.

Contents

#	Exercise
1	Mean, Median, Mode and Range
2	Frequency Distribution
3	Quartiles (Q1, Q2, Q3) and IQR
4	Outlier Detection — IQR Method
5	Skewness (Asymmetry)
6	Continuous Distribution — Uniform Distribution
7	Normal Distribution — Z-Score
8	Normal Distribution — Probability
9	Effect of Outlier Treatment on Mean vs Median
10	Complete EDA on a Student-Mark Dataset

1. Mean, Median, Mode and Range

Problem 1

Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.

Step 1: Calculate the Mean

$$\text{Sum} = 12 + 15 + 18 + 15 + 20 + 22 + 15 + 25 = 142$$

$$\text{Mean} = 142 / 8 = 17.75$$

Step 2: Find the Median

Sorted data: 12, 15, 15, 15, 18, 20, 22, 25 (n = 8, even)

$$\text{Median} = (4\text{th value} + 5\text{th value}) / 2 = (15 + 18) / 2 = 16.5$$

Step 3: Find the Mode

15 occurs 3 times — more often than any other value.

Step 4: Calculate the Range

$$\text{Range} = \text{Maximum} - \text{Minimum} = 25 - 12 = 13$$

Mean : 17.75
Median : 16.5
Mode : 15
Range : 13

The single repeated value (15) pulls the mode below both the mean and the median, while the mean sits slightly above the median — a mild hint of right skew driven by 25.

2. Frequency Distribution

Problem 2

Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.

Step 1: Identify unique values

The values range from 2 to 6. Total observations, n = 12.

Step 2 & 3: Count frequency and compute relative frequency / percentage

Value	Frequency	Relative Frequency	Percentage
2	3	0.2500	25.00%
3	3	0.2500	25.00%
4	3	0.2500	25.00%
5	2	0.1667	16.67%
6	1	0.0833	8.33%
Total	12	1.0000	100%

Conclusion: Values 2, 3 and 4 are equally the most common (3 occurrences each, 25% of the data), while 6 is the least frequent value, appearing only once.

3. Quartiles (Q1, Q2, Q3) and IQR

Problem 3

Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.

Step 1: Find Q2 (median)

There are 10 observations. $Q2 = (5\text{th} + 6\text{th}) / 2 = (15 + 18) / 2 = 16.5$

Step 2: Find Q1

Lower half: 5, 8, 10, 12, 15 → middle value, $Q1 = 10$

Step 3: Find Q3

Upper half: 18, 20, 22, 25, 30 → middle value, $Q3 = 22$

Step 4: Calculate IQR

$IQR = Q3 - Q1 = 22 - 10 = 12$

Q1 : 10
Q2 : 16.5
Q3 : 22
IQR : 12

4. Outlier Detection — IQR Method

Problem 4

Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.

Step 1: Find Q1

Lower half: 10, 12, 13, 15 → $Q1 = (12 + 13) / 2 = 12.5$

Step 2: Find Q3

Upper half: 17, 18, 20, 45 → $Q3 = (18 + 20) / 2 = 19$

Step 3: Calculate IQR

$IQR = Q3 - Q1 = 19 - 12.5 = 6.5$

Step 4: Calculate the lower boundary

Lower Bound = $Q1 - 1.5(IQR) = 12.5 - 1.5(6.5) = 12.5 - 9.75 = 2.75$

Step 5: Calculate the upper boundary

Upper Bound = $Q3 + 1.5(IQR) = 19 + 1.5(6.5) = 19 + 9.75 = 28.75$

Step 6: Identify the outlier

Any value below 2.75 or above 28.75 is an outlier. Since $45 > 28.75$:

45 is an outlier

Step 7: Outlier treatment (median replacement)

Median of the dataset ($n = 9$) is the 5th value = 16.

Treated dataset: 10, 12, 13, 15, 16, 16, 17, 18, 20

As with the worked example, an outlier flagged by the IQR rule should not be deleted automatically — first check whether it is a data-entry error or a genuine extreme value.

5. Skewness (Asymmetry)

Problem 5

Determine the skewness of: 5, 6, 7, 7, 8, 8, 9, 20.

Step 1: Find the Mean

Mean = $(5+6+7+7+8+8+9+20) / 8 = 70 / 8 = 8.75$

Step 2: Find the Median

Sorted ($n = 8$): 5, 6, 7, 7, 8, 8, 9, 20 → Median = $(7 + 8) / 2 = 7.5$

Step 3: Compare Mean and Median

Mean (8.75) > Median (7.5)

Distribution is positively / right skewed

Interpretation: the extreme value 20 pulls the mean well above the median, producing a right-skewed distribution — the same pattern seen in the worked example.

6. Continuous Distribution — Uniform Distribution

Problem 6

A uniformly distributed variable ranges from 10 to 50. Find $P(20 < X < 35)$.

Step 1: Identify limits

$a = 10$, $b = 50$, so we need $P(20 < X < 35)$.

Step 2: Apply the uniform distribution formula

$P(c < X < d) = (d - c) / (b - a)$

$P(20 < X < 35) = (35 - 20) / (50 - 10) = 15 / 40 = 0.375$

$P(20 < X < 35) = 0.375 \rightarrow 37.5\%$ probability

7. Normal Distribution — Z-Score

Problem 7

A normal distribution has mean 100 and standard deviation 15. Find the Z-score for $X = 130$.

Step 1: Formula

$$Z = (X - \mu) / \sigma$$

Step 2: Substitute values

$$Z = (130 - 100) / 15 = 30 / 15 = 2$$

$$Z = 2$$

Interpretation: $X = 130$ lies 2 standard deviations above the mean.

8. Normal Distribution — Probability

Problem 8

For the same normal distribution ($\mu = 100$, $\sigma = 15$), find $P(X < 130)$.

Step 1: Use the Z-score already found

From Exercise 7, $Z = 2$.

Step 2: Look up the standard normal table

$$P(Z < 2) \approx 0.9772$$

$$P(X < 130) = 0.9772 \rightarrow 97.72\%$$

Interpretation: approximately 97.72% of the distribution lies below $X = 130$.

9. Effect of Outlier Treatment on Mean vs Median

Problem 9

Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.

Using the Exercise 4 dataset: 10, 12, 13, 15, 16, 17, 18, 20, 45 (outlier = 45, flagged by the IQR rule).

Step 1: Before treatment

$$\text{Sum} = 166 \rightarrow \text{Mean} = 166 / 9 \approx 18.44$$

$$\text{Sorted, } n = 9 \rightarrow \text{Median (5th value)} = 16$$

Step 2: After treatment (45 replaced by median 16)

Treated dataset: 10, 12, 13, 15, 16, 16, 17, 18, 20

$$\text{Sum} = 137 \rightarrow \text{Mean} = 137 / 9 \approx 15.22$$

$$\text{Median (5th value)} = 16 \text{ (unchanged)}$$

Stage	Mean	Median
Before treatment	18.44	16
After treatment	15.22	16

Conclusion: The mean shifted noticeably ($18.44 \rightarrow 15.22$) once the extreme value was treated, while the median barely moved. This confirms that the mean is highly sensitive to outliers, whereas the median is a robust measure of central tendency.

10. Complete EDA on a Student-Mark Dataset

Problem 10

Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.

Sample dataset (10 students, marks out of 100): 45, 50, 55, 60, 62, 65, 68, 70, 75, 95

Step 1: Central tendency

Sum = 645 $\rightarrow$ Mean = $645 / 10 = 64.5$

Median = $(5\text{th} + 6\text{th}) / 2 = (62 + 65) / 2 = 63.5$

Mode = none (all ten values are unique)

Step 2: Dispersion

Range = $95 - 45 = 50$

Lower half: 45, 50, 55, 60, 62 $\rightarrow$ Q1 = 55

Upper half: 65, 68, 70, 75, 95 $\rightarrow$ Q3 = 70

IQR = $Q3 - Q1 = 70 - 55 = 15$

Population variance ≈ 179.05 , Standard deviation ≈ 13.38

Step 3: Distribution / skewness

Mean (64.5) is slightly greater than the Median (63.5) $\rightarrow$ mild positive / right skew, driven by the high score of 95.

Step 4: Outlier detection (IQR method)

Lower Bound = $55 - 1.5(15) = 32.5$

Upper Bound = $70 + 1.5(15) = 92.5$

$95 > 92.5 \rightarrow$ flagged as an outlier.

```
Mean : 64.5
Median : 63.5
Mode : None
Range : 50
Q1 / Q3 : 55 / 70
IQR : 15
Std. Deviation : 13.38 (approx.)
Skewness : Mild positive (right) skew
Outlier(s) : 95
```

Summary: the class average sits around 64-65 marks, with most students clustered between 55 and 70. The distribution is mildly right-skewed because of one high-scoring student (95), who is statistically an outlier under the IQR rule but most likely represents genuine strong performance rather than a data error — worth reviewing, not automatically discarding.